Nonparametric Statistics
Chapter 3: Nonparametric Regression

Hoang Van Ha
hvha@hcmus.edu.vn

Faculty of Mathematics and Computer Science
University of Science, VNU–HCM

October 6, 2026

Outline

1. Introduction
2. A real example with R
3. The Nadaraya–Watson estimator
4. Properties, risk and bandwidth selection
5. Simulation: Nadaraya–Watson
6. Local polynomial estimators
7. Simulation: local polynomials
8. Exercises

The regression problem

We observe a sample of n pairs

$$(X_1, Y_1), (X_2, Y_2), \ldots, (X_n, Y_n)$$

with the same distribution as a generic pair (X, Y), where $\mathbb{E}[|Y|] < \infty$.

The regression function of Y on X is

$$r(x) \;=\; \mathbb{E}[Y \mid X = x], \qquad x \in \mathbb{R}.$$

We can write the data as

$$Y_i \;=\; r(X_i) + \xi_i, \qquad 1 \le i \le n,$$

where the residuals $\xi_i = Y_i - \mathbb{E}[Y_i \mid X_i]$ satisfy

$$\mathbb{E}[\xi_i \mid X_i] = 0, \qquad \mathbb{V}\mathrm{ar}(\xi_i \mid X_i) = \sigma^2(X_i) < \infty.$$

We write $\sigma^2(x) = \mathbb{V}\mathrm{ar}(Y \mid X = x)$ and, when homoscedastic, $\sigma^2(x) \equiv \sigma^2$.

Parametric vs nonparametric

Parametric regression: we assume $r \in \{r_\theta : \theta \in \Theta \subset \mathbb{R}^p\}$.

Example: linear model $r(x) = \beta_0 + \beta_1 x$.

Nonparametric regression: we only assume

$$r \in \mathcal{F},$$

where $\mathcal{F}$ is an *infinite-dimensional* class of functions (smooth, Hölder, Sobolev, ...).

Goal: estimate r with as few assumptions as possible, and study the risk of the resulting estimator.

Fixed vs random design

The variables $X_1, \ldots, X_n$ form a *design*.

Fixed (deterministic) design. The X_i's are non-random points chosen by the statistician, e.g. $X_i = i/n$. We say we have *fixed effects*.

Random design. The X_i's are i.i.d. from some distribution with density f_X. We say we have *random effects*.

Notation. Throughout, for the homoscedastic case $\sigma^2 = \mathbb{V}\mathrm{ar}(\xi)$, and $f := f_X$ denotes the design density (random design).

Smoothness classes: the Hölder class

The rates of convergence below are stated over *Hölder classes*, which quantify how smooth we assume r to be.

Definition (Hölder class). Let $\beta > 0$ and $L > 0$, and write $\lfloor \beta \rfloor$ for the largest integer $< \beta$. The Hölder class $\mathcal{H}(\beta, L)$ on an interval I is the set of functions $r : I \to \mathbb{R}$ that are $\lfloor \beta \rfloor$ times differentiable and satisfy

$$\left| r^{(\lfloor \beta \rfloor)}(x) - r^{(\lfloor \beta \rfloor)}(x') \right| \leq L |x - x'|^{\beta - \lfloor \beta \rfloor}, \qquad \forall x, x' \in I.$$

Interpretation.

- β is the *degree of smoothness*; larger $\beta \Rightarrow$ easier to estimate.
- $\beta = 1$: r is L-Lipschitz. $\beta = 2$: r' is L-Lipschitz (so r'' exists a.e., bounded).
- The parameter β will control both the optimal bandwidth and the optimal rate.

Risk criteria

To evaluate an estimator $\hat{r}_n$ of r we use:

Pointwise quadratic risk

$$\mathrm{MSE}(x) \;=\; \mathbb{E}\big[(\hat{r}_n(x) - r(x))^2\big] \;=\; \underbrace{(\mathbb{E}\hat{r}_n(x) - r(x))^2}_{b^2(x)\ \text{bias}^2} + \underbrace{\mathbb{V}\mathrm{ar}(\hat{r}_n(x))}_{\text{variance}}.$$

Integrated quadratic risk (MISE)

$$\mathrm{MISE}(\hat{r}_n) \;=\; \mathbb{E}\left[\int (\hat{r}_n(x) - r(x))^2\, dx\right].$$

Bias–variance tradeoff: smoothing too little $\Rightarrow$ small bias, large variance; smoothing too much $\Rightarrow$ small variance, large bias.

A motivating dataset: `mcycle`

Data. The `mcycle` dataset (library `MASS` in R) records accelerometer measurements during a simulated motorcycle crash:

- X = time (ms) after impact,
- Y = head acceleration (g).

The relationship between Y and X is clearly *nonlinear* and not well described by a low-degree polynomial.

A nonparametric estimate $\hat{r}_n(x)$ lets the data speak for itself.

R code: loading the data and plotting

```
library(MASS)
data(mcycle)
plot(mcycle$times, mcycle$accel,
     pch = 4, col = "gray40",
     xlab = "time (ms)",
     ylab = "acceleration (g)",
     main = "mcycle dataset")
```

A few candidate nonparametric estimators (covered in this chapter):

- Regressogram (histogram-type estimator),
- Nadaraya–Watson kernel estimator,
- Local polynomial estimator,
- Projection (orthogonal series) estimator.

R code: a quick Nadaraya–Watson fit

```r
# Method 1: built-in ksmooth
fit <- ksmooth(mcycle$times, mcycle$accel,
               kernel = "normal", bandwidth = 3)

plot(mcycle$times, mcycle$accel, pch = 4, col = "gray40",
     xlab = "time", ylab = "accel")
lines(fit, col = "red", lwd = 2)

# Method 2: np package (with bandwidth selection)
# install.packages("np")
library(np)
bw  <- npregbw(accel ~ times, data = mcycle)
nw  <- npreg(bw)
lines(mcycle$times, fitted(nw), col = "blue", lwd = 2)
```

From histograms to kernels

Idea 1 (regressogram). Partition $[a, b]$ into D bins $I_1, \dots, I_D$ and, on each bin, replace r by the average of the Y_i's with X_i in that bin:

$$\hat{r}_{I,n}(x) \;=\; \sum_{k=1}^{D} \left[\frac{1}{n_k} \sum_{i=1}^{n} Y_i \, \mathbf{1}_{\{X_i \in I_k\}} \right] \mathbf{1}_{I_k}(x), \qquad n_k = \#\{i : X_i \in I_k\}.$$

Drawback: piecewise constant, discontinuous, sensitive to bin placement.

Idea 2. Replace the indicator of a bin by a smooth *kernel* centred at x: this is the Nadaraya–Watson estimator.

Construction (random design)

Assume (X, Y) has joint density f_{XY} and that $f_X(x) > 0$. Then

$$r(x) \;=\; \mathbb{E}[Y \mid X = x] \;=\; \frac{\int y\, f_{XY}(x, y)\, dy}{f_X(x)}.$$

Estimate the numerator and denominator by kernel density estimators:

$$\hat{f}_{n,XY}(x, y) = \frac{1}{n} \sum_{i=1}^{n} K_h(X_i - x)\, K_h(Y_i - y), \qquad \hat{f}_{n,X}(x) = \frac{1}{n} \sum_{i=1}^{n} K_h(X_i - x),$$

with $K_h(\cdot) = \frac{1}{h} K\big(\frac{\cdot}{h}\big)$.

This leads to

$$\hat{r}_n(x) \;=\; \frac{\int y\, \hat{f}_{n,XY}(x, y)\, dy}{\hat{f}_{n,X}(x)}\, \mathbf{1}_{\{\hat{f}_{n,X}(x) \neq 0\}}.$$

The reduction to a weighted average

Proposition. Let $\hat{f}_{n,XY}$ and $\hat{f}_{n,X}$ be the kernel density estimators defined above, where K is a kernel of *order* 1, i.e.

$$\int K(u)\,du = 1, \qquad \int u\,K(u)\,du = 0.$$

Then, whenever $\hat{f}_{n,X}(x) \neq 0$,

$$\frac{\int y\,\hat{f}_{n,XY}(x,y)\,dy}{\hat{f}_{n,X}(x)} \;=\; \frac{\displaystyle\sum_{i=1}^{n} Y_i\,K_h(X_i - x)}{\displaystyle\sum_{i=1}^{n} K_h(X_i - x)}.$$

Remark. If the marginal density f_X is known, one can replace $\hat{f}_{n,X}$ by f_X and obtain the closely related estimator

$$\bar{r}_{n,h}(x) \;=\; \frac{1}{nh\,f_X(x)} \sum_{i=1}^{n} Y_i\,K\big(\tfrac{X_i - x}{h}\big)\,.$$

Proof of the Proposition

It suffices to simplify the numerator. By definition of $\hat{f}_{n,XY}$,

$$\int y\, \hat{f}_{n,XY}(x,y)\, dy = \frac{1}{n}\sum_{i=1}^{n} K_h(X_i - x) \int y\, K_h(Y_i - y)\, dy.$$

Fix i and compute the inner integral with the change of variable $u = (Y_i - y)/h$, so that $y = Y_i - hu$ and $dy = -h\, du$:

$$\int y\, K_h(Y_i - y)\, dy = \int (Y_i - hu)\, K(u)\, du = Y_i \underbrace{\int K(u)\, du}_{=1} - h \underbrace{\int u\, K(u)\, du}_{=0} = Y_i.$$

Here we used precisely that K has *order* 1.

Therefore $\int y\, \hat{f}_{n,XY}(x,y)\, dy = \frac{1}{n}\sum_{i=1}^{n} Y_i\, K_h(X_i - x)$, and dividing by $\hat{f}_{n,X}(x) = \frac{1}{n}\sum_i K_h(X_i - x)$ gives the claim. ∎

The Nadaraya–Watson estimator

Corollary. If K is a kernel of order 1, the estimator above reduces to

$$\hat{r}_n^{\mathrm{NW}}(x) = \frac{\sum_{i=1}^n Y_i K_h(X_i - x)}{\sum_{i=1}^n K_h(X_i - x)} \mathbf{1}_{\left\{\sum_i K_h(X_i - x) \neq 0\right\}}, \qquad x \in \mathbb{R}.$$

Common kernels:

- Rectangular: $K(u) = \frac{1}{2}\mathbf{1}_{\{|u| \leq 1\}}$,
- Epanechnikov: $K(u) = \frac{3}{4}(1 - u^2)_+$,
- Gaussian: $K(u) = \frac{1}{\sqrt{2\pi}} e^{-u^2/2}$.

R: `ksmooth(...)` or `npreg(...)` in package `np`.

Weighted-mean form

The Nadaraya–Watson estimator is a weighted mean of the Y_i:

$$\hat{r}_n^{\mathrm{NW}}(x) \;=\; \sum_{i=1}^{n} Y_i \, W_{n,i}(x),$$

with weights

$$W_{n,i}(x) \;=\; \frac{K_h(X_i - x)}{\sum_{j=1}^{n} K_h(X_j - x)} \, \mathbf{1}_{\left\{\sum_j K_h(X_j - x) \neq 0\right\}}.$$

The weights depend only on $X_1, \ldots, X_n$ and satisfy

$$\sum_{i=1}^{n} W_{n,i}(x) \;=\; 1.$$

$\hat{r}_n^{\mathrm{NW}}$ is a *linear* estimator in $(Y_1, \ldots, Y_n)$ (not to be confused with linear regression!).

Linear nonparametric regression estimators

Definition. An estimator $\hat{r}_n(x)$ of $r(x)$ is called a *linear nonparametric regression estimator* if it can be written as

$$\hat{r}_n(x) \;=\; \sum_{i=1}^{n} Y_i \, W_{n,i}(x),$$

where the weights $W_{n,i}(x) = W_{n,i}(x; X_1, \ldots, X_n)$ depend only on n, i, x and the design points $X_1, \ldots, X_n$ (not on the Y_j's).

Typically, the weights of a linear regression estimator satisfy

$$\sum_{i=1}^{n} W_{n,i}(x) \;=\; 1 \qquad \text{for (almost) all } x.$$

Examples.

- Regressogram (histogram-type),
- Nadaraya–Watson,
- local polynomial estimator $\hat{r}_n^{\mathrm{LP}(\ell)}$,
- projection / orthogonal series estimator.

Why "linear"?

Linearity is in the responses Y_i, not in x.

Three useful consequences:

- In matrix form on the design points, $\hat{\mathbf{y}} = \mathbf{W}\,\mathbf{y}$ for some smoothing matrix $\mathbf{W}$ depending only on the X_i's.
- Conditional on the design, bias and variance decompose cleanly:

$$\mathbb{E}[\hat{r}_n(x) \mid \mathbf{X}] = \sum_{i=1}^{n} W_{n,i}(x)\, r(X_i), \qquad \mathbb{V}\mathrm{ar}(\hat{r}_n(x) \mid \mathbf{X}) = \sum_{i=1}^{n} \sigma^2(X_i)\, W_{n,i}^2(x).$$

  (Homoscedastic case: $\mathbb{V}\mathrm{ar} = \sigma^2 \sum_i W_{n,i}^2(x)$.)

- Closed-form leave-one-out cross-validation: $Y_i - \hat{r}_n^{(-i)}(X_i) = \dfrac{Y_i - \hat{r}_n(X_i)}{1 - W_{ii}}$.

Matrix form

At the observed design points (x_i, y_i), $1 \le i \le n$, set $\hat{y}_i = \hat{r}_n^{\mathrm{NW}}(x_i)$ and let $\mathbf{W}$ be the $n \times n$ matrix with entries $w_{ij} = W_{n,j}(x_i)$. Then

$$\hat{\mathbf{y}} \;=\; \mathbf{W}\,\mathbf{y}.$$

The trace

$$\tau \;=\; \mathrm{tr}(\mathbf{W}) \;=\; \sum_{i=1}^{n} W_{n,i}(x_i)$$

plays the role of an *effective number of parameters* (degrees of freedom) of the smoother.

Small $h \Rightarrow \tau$ close to n (interpolation). Large $h \Rightarrow \tau$ close to 1 (over-smoothing).

Bias and variance: fixed design

Fixed design $X_i = i/n$ on $[0,1]$, r twice differentiable. Under standard kernel assumptions (K symmetric, order 1, $\mu_2(K) := \int u^2 K(u)\,du < \infty$, $R(K) := \int K^2(u)\,du < \infty$), for x in the interior of $[0,1]$,

$$\mathbb{E}\big[\hat{r}_n^{\mathrm{NW}}(x)\big] - r(x) \;=\; \frac{h^2}{2}\, r''(x)\,\mu_2(K) \;+\; o(h^2),$$

$$\mathbb{V}\mathrm{ar}\big(\hat{r}_n^{\mathrm{NW}}(x)\big) \;=\; \frac{\sigma^2}{nh}\, R(K) \;+\; o\big(\tfrac{1}{nh}\big)\,.$$

Boundary effect: near 0 or 1, the bias is of order h instead of h^2 (the kernel sees fewer data points). Local polynomial estimators fix this issue.

Bias and variance: random design

Now $X_i \overset{iid}{\sim} f$, with $f(x) > 0$ and $f, r \in C^2$ near x. Conditionally on the design, a leading-order (stochastic linearisation) expansion gives

$$\boxed{\mathbb{E}[\hat{r}_n^{\mathrm{NW}}(x) \mid \mathbf{X}] - r(x) \;\approx\; h^2 \mu_2(K) \left[\tfrac{1}{2} r''(x) \;+\; \frac{r'(x)\, f'(x)}{f(x)} \right],}$$

$$\mathbb{V}\mathrm{ar}[\hat{r}_n^{\mathrm{NW}}(x) \mid \mathbf{X}] \;\approx\; \frac{\sigma^2(x)}{nh\, f(x)} R(K).$$

Key observation. The bias carries a *design-dependent* term $\dfrac{r'f'}{f}$: Nadaraya–Watson is biased wherever the data are unevenly spread ($f' \neq 0$), even if $r'' = 0$. This is the defect local linear regression removes.

Random design: sketch of the bias computation

Write $\hat{r}_n^{\mathrm{NW}}(x) = \hat{a}(x)/\hat{b}(x)$ with $\hat{a} = \frac{1}{n}\sum_i Y_i K_h(X_i - x)$, $\hat{b} = \hat{f}_{n,X}(x)$. Using $\mathbb{E}[g(X)K_h(X-x)] = \int g(x+hv)K(v)f(x+hv)\,dv$ and a Taylor expansion,

$$\mathbb{E}\hat{b} = f(x) + \tfrac{h^2}{2}\mu_2 f''(x) + o(h^2), \quad \mathbb{E}\hat{a} = (rf)(x) + \tfrac{h^2}{2}\mu_2 (rf)''(x) + o(h^2).$$

Since $r(x) = \mathbb{E}Y$ |- consistency gives $\hat{r}_n^{\mathrm{NW}}(x) - r(x) \approx (\hat{a} - r(x)\hat{b})/f(x)$, and

$$\mathbb{E}[\hat{a} - r\hat{b}] \approx \tfrac{h^2}{2}\mu_2\big[(rf)'' - rf''\big] = \tfrac{h^2}{2}\mu_2\big[r''f + 2r'f'\big],$$

using $(rf)'' = r''f + 2r'f' + rf''$. Dividing by $f(x)$ yields the boxed bias. ■

Variance: to leading order $\mathbb{V}\mathrm{ar}(\hat{a} - r\hat{b} \mid \mathbf{X}) \approx \frac{\sigma^2(x)f(x)}{nh}R(K)$, and dividing by $f(x)^2$ gives $\frac{\sigma^2(x)R(K)}{nhf(x)}$.

Pointwise MSE and the optimal bandwidth

Putting the two terms together (homoscedastic, interior x),

$$\mathrm{MSE}(x) \;=\; b^2(x) + \mathbb{V}\mathrm{ar}(\hat{r}_n^{\mathrm{NW}}(x)) \;\le\; C_1\, h^4 \;+\; \frac{C_2}{nh}.$$

Minimising in h gives the optimal pointwise bandwidth

$$h^* \;\asymp\; n^{-1/5}, \qquad \mathrm{MSE}(x) \;\asymp\; n^{-4/5}.$$

Slower than the parametric rate n^{-1}, but *nothing* was assumed about the parametric form of r.

More generally, if $r \in \mathcal{H}(\beta, L)$ and the smoothing method can exploit that smoothness, the optimal rate is $\mathrm{MSE} \asymp n^{-2\beta/(2\beta+1)}$.

Bandwidth selection: leave-one-out cross-validation

Let $\hat{r}_h^{(-i)}$ be the estimator obtained by removing (X_i, Y_i) from the sample. The *leave-one-out CV* criterion is

$$\mathrm{CV}(h) \;=\; \frac{1}{n}\sum_{i=1}^{n}\big(Y_i - \hat{r}_h^{(-i)}(X_i)\big)^2.$$

For linear smoothers $\hat{\mathbf{y}} = \mathbf{W}\mathbf{y}$ one can show

$$\mathrm{CV}(h) \;=\; \frac{1}{n}\sum_{i=1}^{n}\left(\frac{Y_i - \hat{r}_h(X_i)}{1 - W_{ii}}\right)^2$$

which avoids fitting n separate estimators.

We choose $\hat{h} \;=\; \underset{h\in\mathcal{H}}{\arg\min}\,\mathrm{CV}(h)$.

Generalised cross-validation (GCV)

If we replace each W_{ii} by its average $\bar{W} = \frac{1}{n}\sum_{i=1}^{n} W_{ii} = \tau/n$ we obtain

$$\mathrm{GCV}(h) \;=\; \frac{1}{(1-\tau/n)^2}\,\sum_{i=1}^{n}\big(Y_i - \hat{r}_h(X_i)\big)^2.$$

Advantages of GCV:

- invariant under rotations of $\mathbf{y}$,
- often more stable than CV in finite samples,
- only one matrix trace to compute.

Both CV and GCV are implemented in standard R packages (`np`, `locfit`, `KernSmooth`).

Simulation setup

We simulate from

$$Y_i \;=\; r(X_i) + \xi_i, \qquad r(x) = \sin(2\pi x) + 0.5\,x, \qquad \xi_i \sim \mathcal{N}(0, 0.25^2),$$

with X_i uniform on $[0, 1]$ and $n = 200$.

```
set.seed(2025)
n  <- 200
X  <- sort(runif(n))
r  <- function(x) sin(2*pi*x) + 0.5*x
Y  <- r(X) + rnorm(n, sd = 0.25)
plot(X, Y, pch = 19, col = "gray60",
     xlab = "x", ylab = "y")
xx <- seq(0, 1, length = 400)
lines(xx, r(xx), col = "black", lwd = 2)  # true function
```

Effect of the bandwidth h

```
# three different bandwidths
h.small <- 0.02
h.opt   <- 0.08
h.big   <- 0.30

fit1 <- ksmooth(X, Y, "normal", h.small, x.points = xx)
fit2 <- ksmooth(X, Y, "normal", h.opt,   x.points = xx)
fit3 <- ksmooth(X, Y, "normal", h.big,   x.points = xx)

plot(X, Y, pch = 19, col = "gray70")
lines(xx, r(xx),  lwd = 2)
lines(fit1, col = "red",   lwd = 2)   # under-smoothing
lines(fit2, col = "blue",  lwd = 2)   # nearly optimal
lines(fit3, col = "green", lwd = 2)   # over-smoothing
legend("topright",
   c("true r","h small","h optimal","h large"),
   col = c("black","red","blue","green"), lwd = 2)
```

Bandwidth selection by cross-validation

```r
# leave-one-out CV for Nadaraya-Watson (Gaussian kernel)
NW <- function(x, X, Y, h){
  K <- dnorm((X - x)/h)
  sum(K * Y) / sum(K)
}

CV <- function(h){
  yhat <- sapply(X, function(xi){
    sel <- X != xi
    NW(xi, X[sel], Y[sel], h)
  })
  mean((Y - yhat)^2)
}

h.grid <- seq(0.01, 0.30, length = 40)
cv.val <- sapply(h.grid, CV)
h.cv   <- h.grid[which.min(cv.val)]
plot(h.grid, cv.val, type = "b",
     xlab = "h", ylab = "CV(h)")
abline(v = h.cv, col = "red", lty = 2)
```

Motivation

The Nadaraya–Watson estimator can be rewritten as a *local constant* fit:

$$\hat{r}_n^{\mathrm{NW}}(x) \;=\; \arg\min_{\theta\in\mathbb{R}} \sum_{i=1}^{n} K_h(X_i - x)\,(Y_i - \theta)^2.$$

(Set the derivative in θ to zero: $\sum_i K_h(X_i - x)(Y_i - \theta) = 0$, which solves to the NW ratio.)

Idea. Replace the constant θ by a *polynomial* of degree ℓ in $(X_i - x)$.

If r is ℓ times differentiable, Taylor's formula gives, for u near x,

$$r(u) \;=\; r(x) + r'(x)(u-x) + \cdots + \frac{r^{(\ell)}(x)}{\ell!}\,(u-x)^{\ell} + R_\ell.$$

This suggests fitting a polynomial *locally* around x.

Rescaled Taylor expansion

Introducing the bandwidth h, write

$$r(u) \;=\; r(x) + r'(x)\,h\left(\frac{u-x}{h}\right) + \cdots + \frac{r^{(\ell)}(x)\,h^{\ell}}{\ell!}\left(\frac{u-x}{h}\right)^{\ell} \;=\; \theta(x)^{\top} V_{\ell}\left(\frac{u-x}{h}\right),$$

with

$$\theta(x) \;=\; \big(r(x),\, r'(x)\,h,\, \ldots,\, r^{(\ell)}(x)\,h^{\ell}\big)^{\top}, \qquad V_{\ell}(z) \;=\; \big(1,\, z,\, \tfrac{z^2}{2!},\, \ldots,\, \tfrac{z^{\ell}}{\ell!}\big)^{\top}.$$

The coefficients of $\theta(x)$ encode $r(x)$ *and* its first ℓ derivatives at x.

Definition

Definition. Given a positive kernel K, a bandwidth $h > 0$ and an integer $\ell \geq 0$, the local polynomial estimator of order ℓ is

$$\hat{\theta}_n(x) \;=\; \underset{\theta \in \mathbb{R}^{\ell+1}}{\arg\min} \sum_{i=1}^{n} K\left(\frac{X_i - x}{h}\right) \left[\, Y_i - \theta^\top V_\ell\left(\tfrac{X_i - x}{h}\right)\right]^2,$$

and the estimator of $r(x)$ is the first coordinate of $\hat{\theta}_n(x)$:

$$\hat{r}_n^{\mathrm{LP}(\ell)}(x) \;=\; \hat{\theta}_n(x)^\top V_\ell(0) \;=\; \left[\hat{\theta}_n(x)\right]_1.$$

Special cases.

- $\ell = 0$: $\hat{r}_n^{\mathrm{LP}(\ell)}[0] = \hat{r}_n^{\mathrm{NW}}$ (Nadaraya–Watson).
- $\ell = 1$: *local linear* estimator (the most popular choice).
- $\ell = 2$: local quadratic estimator.

The normal equations

Work with the *raw-monomial* basis $U_i = \big(1,\, (X_i - x),\, \ldots,\, (X_i - x)^\ell\big)^\top$; the fitted value $\hat{r}_n^{\mathrm{LP}(\ell)}(x) = [\hat{\theta}]_1$ is unchanged by this re-parametrisation (the first basis function is still the constant 1). The objective is

$$\Phi(\theta) \;=\; \sum_{i=1}^{n} K_h(X_i - x)\,\big(Y_i - \theta^\top U_i\big)^2.$$

Since Φ is a (convex) quadratic in θ, the minimiser solves $\nabla_\theta \Phi(\theta) = \mathbf{0}$:

$$\nabla_\theta \Phi(\theta) = -2 \sum_{i=1}^{n} K_h(X_i - x)\, U_i \big(Y_i - \theta^\top U_i\big) = \mathbf{0}.$$

Rearranging gives the normal equations

$$\underbrace{\Big(\textstyle\sum_i K_h(X_i - x)\, U_i U_i^\top\Big)}_{\mathbf{B}_n(x)}\, \hat{\theta} \;=\; \underbrace{\textstyle\sum_i K_h(X_i - x)\, U_i\, Y_i}_{\mathbf{a}_n(x)}, \qquad \hat{\theta} = \mathbf{B}_n(x)^{-1}\mathbf{a}_n(x).$$

The moment matrix: a Hankel structure

With the raw-monomial basis, the entries of $\mathbf{B}_n(x) = \sum_i K_h(X_i - x) U_i U_i^\top$ are *weighted design moments*. Define

$$S_{n,k}(x) \;=\; \sum_{i=1}^{n} K_h(X_i - x)\,(X_i - x)^k, \qquad T_{n,k}(x) \;=\; \sum_{i=1}^{n} K_h(X_i - x)\,(X_i - x)^k\, Y_i.$$

Then $[\mathbf{B}_n(x)]_{j,k} = S_{n,j+k}(x)$ and $[\mathbf{a}_n(x)]_j = T_{n,j}(x)$ $(0 \le j, k \le \ell)$:

$$\mathbf{B}_n(x) = \begin{pmatrix} S_{n,0} & S_{n,1} & \cdots & S_{n,\ell} \\ S_{n,1} & S_{n,2} & \cdots & S_{n,\ell+1} \\ \vdots & \vdots & \ddots & \vdots \\ S_{n,\ell} & S_{n,\ell+1} & \cdots & S_{n,2\ell} \end{pmatrix}, \qquad \mathbf{a}_n(x) = \begin{pmatrix} T_{n,0} \\ T_{n,1} \\ \vdots \\ T_{n,\ell} \end{pmatrix}.$$

$\mathbf{B}_n$ is a Hankel matrix built from $S_{n,0}, \ldots, S_{n,2\ell}$, and $\hat{r}_n^{\mathrm{LP}(\ell)}(x) = e_1^\top \mathbf{B}_n(x)^{-1} \mathbf{a}_n(x)$, where $e_1 = (1, 0, \ldots, 0)^\top$.

Matrix form (design matrix)

Equivalently, let $\mathbf{y} = (Y_1, \ldots, Y_n)^\top$, $\mathbf{W}(x) = \operatorname{diag}\big(K_h(X_1 - x), \ldots, K_h(X_n - x)\big)$ and

$$\mathbf{V}(x) = \begin{pmatrix} 1 & (X_1 - x) & \cdots & (X_1 - x)^\ell \\ \vdots & \vdots & & \vdots \\ 1 & (X_n - x) & \cdots & (X_n - x)^\ell \end{pmatrix} \quad (n \times (\ell + 1)).$$

The problem is the *weighted least squares* problem

$$\min_\theta \big(\mathbf{y} - \mathbf{V}\theta\big)^\top \mathbf{W}\big(\mathbf{y} - \mathbf{V}\theta\big), \qquad \mathbf{B}_n = \mathbf{V}^\top \mathbf{W}\mathbf{V}, \quad \mathbf{a}_n = \mathbf{V}^\top \mathbf{W}\mathbf{y},$$

with solution

$$\hat\theta(x) = \big(\mathbf{V}^\top \mathbf{W}\mathbf{V}\big)^{-1} \mathbf{V}^\top \mathbf{W}\,\mathbf{y}, \qquad \hat r_n^{\mathrm{LP}(\ell)}(x) = e_1^\top \hat\theta(x) = \boldsymbol{\ell}(x)^\top \mathbf{y},$$

where the effective weights are $\boldsymbol{\ell}(x)^\top = e_1^\top \big(\mathbf{V}^\top \mathbf{W}\mathbf{V}\big)^{-1} \mathbf{V}^\top \mathbf{W}$. This exhibits $\hat r_n^{\mathrm{LP}(\ell)}$ as a *linear* smoother.

Local linear estimator in closed form ($\ell = 1$)

For $\ell = 1$ the normal equations are a 2×2 system:

$$\begin{pmatrix} S_{n,0} & S_{n,1} \\ S_{n,1} & S_{n,2} \end{pmatrix} \begin{pmatrix} \hat{\theta}_0 \\ \hat{\theta}_1 \end{pmatrix} = \begin{pmatrix} T_{n,0} \\ T_{n,1} \end{pmatrix}.$$

Inverting the 2×2 matrix (determinant $S_{n,0}S_{n,2} - S_{n,1}^2$) and taking the first coordinate,

$$\boxed{\hat{r}_n^{\mathrm{LL}}(x) = \hat{\theta}_0 = \frac{S_{n,2}\, T_{n,0} - S_{n,1}\, T_{n,1}}{S_{n,0}S_{n,2} - S_{n,1}^2} = \frac{\sum_{i=1}^n w_i(x)\, Y_i}{\sum_{i=1}^n w_i(x)},}$$

with

$$w_i(x) = K_h(X_i - x)\big[\, S_{n,2}(x) - (X_i - x)\, S_{n,1}(x) \,\big].$$

Note. $\sum_i w_i(x) = S_{n,0}S_{n,2} - S_{n,1}^2$, so the weights $\ell_i(x) = w_i(x) / \sum_j w_j(x)$ sum to 1 by construction.

The key algebraic property: polynomial reproduction

Lemma (discrete moments). Let $\ell_i(x)$ be the local-polynomial-of-order-ℓ weights ($\hat{r}_n^{\mathrm{LP}(\ell)}(x) = \sum_i \ell_i(x) Y_i$), and assume $\mathbf{B}_n(x)$ is invertible. Then

$$\boxed{\sum_{i=1}^{n} \ell_i(x)\,(X_i - x)^k \;=\; \delta_{0,k} \qquad \text{for } k = 0, 1, \ldots, \ell.}$$

Equivalently, $\hat{r}_n^{\mathrm{LP}(\ell)}$ *reproduces polynomials of degree* $\leq \ell$: if $Y_i = P(X_i)$ for a polynomial P with $\deg P \leq \ell$, then $\hat{r}_n^{\mathrm{LP}(\ell)}(x) = P(x)$ exactly.

Consequence. For local linear ($\ell = 1$): $\sum_i \ell_i(x) = 1$ *and* $\sum_i \ell_i(x)(X_i - x) = 0$. The second identity is what the NW weights fail to satisfy in general — it is the source of the improved bias.

Proof of the reproduction lemma

The weight vector is $\ell(x)^\top = e_1^\top \mathbf{B}_n^{-1} \mathbf{V}^\top \mathbf{W}$. Let $\mathbf{c}_k \in \mathbb{R}^n$ have entries $(X_i - x)^k$. Then

$$\sum_{i=1}^{n} \ell_i(x)(X_i - x)^k = \ell(x)^\top \mathbf{c}_k = e_1^\top \mathbf{B}_n^{-1} \mathbf{V}^\top \mathbf{W}\, \mathbf{c}_k.$$

Now compute the j-th entry of $\mathbf{V}^\top \mathbf{W} \mathbf{c}_k$ $(0 \le j \le \ell)$:

$$[\mathbf{V}^\top \mathbf{W} \mathbf{c}_k]_j = \sum_{i=1}^{n} (X_i - x)^j \, K_h(X_i - x) \, (X_i - x)^k = S_{n,j+k}(x) = [\mathbf{B}_n]_{j,k}.$$

Hence $\mathbf{V}^\top \mathbf{W} \mathbf{c}_k = \mathbf{B}_n \, e_k$, the $(k{+}1)$-th column of $\mathbf{B}_n$ (for $k \le \ell$). Therefore

$$\sum_{i=1}^{n} \ell_i(x)(X_i - x)^k = e_1^\top \mathbf{B}_n^{-1} \mathbf{B}_n \, e_k = e_1^\top e_k = \delta_{0,k}.$$

The polynomial-reproduction statement follows by linearity of $Y_i \mapsto \sum_i \ell_i Y_i$ and a Taylor/monomial expansion of P. ■

Conditional bias of local linear: design adaptivity

Condition on the design. By linearity and $\sum_i \ell_i = 1$,

$$\mathbb{E}[\hat{r}_n^{\mathrm{LL}}(x) \mid \mathbf{X}] - r(x) = \sum_{i=1}^{n} \ell_i(x)\, r(X_i) - r(x) = \sum_{i=1}^{n} \ell_i(x)\big[r(X_i) - r(x)\big].$$

Taylor-expand $r(X_i) = r(x) + r'(x)(X_i - x) + \frac{1}{2} r''(x)(X_i - x)^2 + o(\cdot)$ and use the two discrete-moment identities $\sum_i \ell_i = 1$, $\sum_i \ell_i (X_i - x) = 0$:

$$\mathbb{E}[\hat{r}_n^{\mathrm{LL}}(x) \mid \mathbf{X}] - r(x) = \tfrac{1}{2} r''(x) \sum_{i=1}^{n} \ell_i(x)(X_i - x)^2 + o(h^2) \;\approx\; \boxed{\tfrac{1}{2}\, h^2 \mu_2(K)\, r''(x).}$$

No f'/f term. The leading bias depends only on r'' — not on the design density or its derivative. This is *design adaptation* (Fan, 1992/93): the same leading bias holds in the interior *and*, to leading order, at the boundary.

Conditional variance of local linear

By linearity, $\mathbb{V}\mathrm{ar}[\hat{r}_n^{\mathrm{LL}}(x) \mid \mathbf{X}] = \sum_{i=1}^n \ell_i(x)^2\,\sigma^2(X_i)$. To leading order the squared effective weights behave like $\ell_i(x)^2 \approx \frac{1}{(nhf(x))^2} K^2\big(\frac{X_i - x}{h}\big)$, so

$$\mathbb{V}\mathrm{ar}[\hat{r}_n^{\mathrm{LL}}(x) \mid \mathbf{X}] \;\approx\; \frac{\sigma^2(x)\,R(K)}{nh\,f(x)}, \qquad R(K) = \int K^2(u)\,du.$$

Summary (local linear, interior *and* boundary):

$$\mathrm{bias} \approx \tfrac{1}{2} h^2 \mu_2(K)\, r''(x), \qquad \mathrm{var} \approx \frac{\sigma^2(x) R(K)}{nhf(x)}.$$

Same variance as NW, strictly simpler (design-free) bias.

Nadaraya–Watson vs local linear: the comparison

	Nadaraya–Watson ($\ell = 0$)	**Local linear** ($\ell = 1$)
Leading bias	$h^2\mu_2\left[\frac{r''}{2} + \frac{r'f'}{f}\right]$	$\frac{h^2\mu_2}{2}\, r''$
Leading variance	$\frac{\sigma^2 R(K)}{nhf}$	$\frac{\sigma^2 R(K)}{nhf}$
Boundary bias	$O(h)$ (degrades)	$O(h^2)$ (no degradation)
Design term $r'f'/f$	present	absent

Takeaway. Local linear has the same asymptotic variance as NW but a smaller, design-free bias and no boundary degradation — hence the recommendation to use $\ell = 1$ as the default.

Asymptotic equivalent kernel

Rescaling the moments, $S_{n,k}(x) \approx nh^k f(x)\,\mu_k(K)$ with $\mu_k = \int u^k K(u)\,du$. Writing $z_i = (X_i - x)/h$, the effective weights take the form

$$\ell_i(x) \;\approx\; \frac{1}{nh\,f(x)}\; K_\ell^*(z_i), \qquad K_\ell^*(u) \;=\; e_1^\top \mathbf{S}^{-1}\left(1, u, \ldots, u^\ell\right)^\top K(u),$$

where $\mathbf{S} = (\mu_{j+k})_{0 \le j,k \le \ell}$ is the limiting moment matrix.

The equivalent kernel K_ℓ^* inherits the discrete moments:

$$\int K_\ell^*(u)\,du = 1, \qquad \int u^k K_\ell^*(u)\,du = 0 \quad (k = 1, \ldots, \ell).$$

So K_ℓ^* is automatically a *higher-order kernel*: local polynomials build bias cancellation *into the fit*, instead of requiring a hand-designed higher-order K. For $\ell = 1$ and symmetric K one gets $K_1^* = K$ in the interior.

Properties

- $\hat{r}_n^{\mathrm{LP}(\ell)}$ is a linear estimator in $(Y_1, \ldots, Y_n)$.
- It simultaneously estimates $r(x)$ *and* its derivatives $r'(x), \ldots, r^{(\ell)}(x)$ via $[\hat{\theta}_n(x)]_{k+1} = \hat{r}^{(k)}(x)\, h^k / k!$ (in the V_ℓ parametrisation).
- It does *not* suffer from boundary bias: near 0 or 1, the local polynomial estimator of order $\ell \geq 1$ keeps the same rate as in the interior.
- Bias and variance, assuming $r \in \mathcal{H}(\beta, L)$ with $\ell \geq \lfloor \beta \rfloor$:

$$|b(x)| \;\leq\; C_1\, L\, h^{\beta}, \qquad \mathbb{V}\mathrm{ar}(\hat{r}_n^{\mathrm{LP}(\ell)}(x) \mid \mathbf{X}) \;\leq\; \frac{C_2\, \sigma^2}{nh}.$$

Bias–variance tradeoff and optimal rate

For $r \in \mathcal{H}(\beta, L)$ with $\ell \geq \lfloor \beta \rfloor$,

$$\mathrm{MSE}(x) \;\leq\; C_1\, h^{2\beta} \;+\; \frac{C_2}{nh}.$$

Choosing $h^* \;=\; c\, n^{-1/(2\beta+1)}$, $c > 0$, yields

$$\limsup_{n\to\infty} \sup_{r\in\mathcal{H}(\beta,L)} \sup_{x\in[0,1]} \mathbb{E}\Big[\, n^{2\beta/(2\beta+1)} \big|\hat{r}^*(x) - r(x)\big|^2 \Big] \;\leq\; C \;<\; \infty.$$

This is the minimax-optimal rate over Hölder classes: local polynomials are rate-optimal.

Local linear fit by hand (via the $S_{n,k}$ moments)

```
loc.lin <- function(x, X, Y, h){
  K  <- dnorm((X - x)/h)              # Gaussian weights
  S0 <- sum(K);           S1 <- sum(K * (X - x))
  S2 <- sum(K * (X - x)^2)
  T0 <- sum(K * Y);       T1 <- sum(K * (X - x) * Y)
  (S2 * T0 - S1 * T1) / (S0 * S2 - S1^2)   # theta_0 = r_hat(x)
}

xx   <- seq(0, 1, length = 400)
yhat <- sapply(xx, function(x0) loc.lin(x0, X, Y, h = 0.08))

plot(X, Y, pch = 19, col = "gray70")
lines(xx, r(xx),  lwd = 2)
lines(xx, yhat,   col = "purple", lwd = 2)
```

The code is a line-by-line transcription of the closed-form $\hat{r}_n^{\mathrm{LL}}(x) = (S_2 T_0 - S_1 T_1)/(S_0 S_2 - S_1^2)$.

Using the locfit / KernSmooth packages

```
# package KernSmooth: local polynomial of given degree
library(KernSmooth)
fit.ll <- locpoly(X, Y, degree = 1,
                  bandwidth = 0.08, gridsize = 400)
fit.lq <- locpoly(X, Y, degree = 2,
                  bandwidth = 0.08, gridsize = 400)

plot(X, Y, pch = 19, col = "gray70")
lines(xx, r(xx), lwd = 2)
lines(fit.ll, col = "blue",   lwd = 2)   # local linear
lines(fit.lq, col = "orange", lwd = 2)   # local quadratic
legend("topright",
       c("true r","local linear","local quadratic"),
       col = c("black","blue","orange"), lwd = 2)
```

Boundary bias: Nadaraya–Watson vs local linear

```r
# Compare fits near the boundary x = 0 and x = 1
xx  <- seq(0, 1, length = 400)
nw  <- ksmooth(X, Y, "normal", bandwidth = 0.10,
               x.points = xx)
ll  <- sapply(xx, function(x0) loc.lin(x0, X, Y, h = 0.05))

plot(X, Y, pch = 19, col = "gray80")
lines(xx, r(xx),    lwd = 2)              # truth
lines(nw,  col = "red",   lwd = 2)        # NW: bends at edges
lines(xx, ll, col = "blue", lwd = 2)      # local linear
legend("top", c("true r","NW","local linear"),
       col = c("black","red","blue"), lwd = 2)
# Observe: NW is visibly biased toward the interior mean
# near x=0 and x=1; local linear tracks r much better.
```

What we observe in the simulations

- Nadaraya–Watson and local linear estimators give very similar fits in the interior of $[0, 1]$ (same leading variance, both $O(h^2)$ bias there).
- Near the boundaries $x \approx 0$ and $x \approx 1$, local linear clearly outperforms Nadaraya–Watson: its bias stays $O(h^2)$ while NW degrades to $O(h)$.
- Local quadratic captures peaks and valleys more sharply, at the price of increased variance.
- As n increases, the optimal bandwidth shrinks like $n^{-1/5}$ (for $r \in \mathcal{H}(2, L)$), and the MSE decreases like $n^{-4/5}$.

Practical rule of thumb: local linear is a safe default; use $\ell = 2$ when estimating derivatives.

Exercises (1/3)

Exercise 1. Show that, for a positive kernel K,

$$\hat{r}_n^{\mathrm{NW}}(x) \;=\; \arg\min_{\theta \in \mathbb{R}} \sum_{i=1}^{n} K_h(X_i - x)\,(Y_i - \theta)^2.$$

Exercise 2. Let $\hat{r}_n^{\mathrm{NW}}$ be the Nadaraya–Watson estimator with weights $W_{n,i}(x)$. Prove that

$$\sum_{i=1}^{n} W_{n,i}(x) \;=\; 1 \qquad \text{whenever } \sum_{j=1}^{n} K_h(X_j - x) \neq 0.$$

Deduce that $\hat{r}_n^{\mathrm{NW}}$ reproduces constants: if $Y_i = c$ for all i, then $\hat{r}_n^{\mathrm{NW}}(x) = c$ for all x in the support.

Exercises (2/3)

Exercise 3 (fixed design). Assume $X_i = i/n$, $1 \le i \le n$, and $\xi_i \sim \mathcal{N}(0, \sigma^2)$ i.i.d. Consider

$$\hat{r}_n(x) \;=\; \frac{1}{n} \sum_{i=1}^{n} Y_i \, K_h(X_i - x) \, \mathbf{1}_{[0,1]}(x).$$

(a) Compute $\mathbb{E}[\hat{r}_n(x)]$ and give an expansion of the bias as $n \to \infty$, $h \to 0$, $nh \to \infty$.

(b) Compute $\mathbb{V}\mathrm{ar}(\hat{r}_n(x))$ and find its leading order.

(c) Deduce the optimal bandwidth h^* minimising the MSE, and the corresponding rate.

Exercise 4 (normal equations $\Rightarrow$ local linear). Starting from the 2×2 normal equations $\mathbf{B}_n \hat{\theta} = \mathbf{a}_n$ with $[\mathbf{B}_n]_{jk} = S_{n,j+k}$, derive the closed form

$$\hat{r}_n^{\mathrm{LL}}(x) = \frac{\sum_i w_i(x) Y_i}{\sum_i w_i(x)}, \quad w_i(x) = K_h(X_i - x)\big[S_{n,2} - (X_i - x) S_{n,1}\big].$$

Exercises (3/3)

Exercise 5 (polynomial reproduction). Using the identity $\mathbf{V}^\top \mathbf{W}\mathbf{c}_k = \mathbf{B}_n\, e_k$ for $k \le \ell$, prove that the order-ℓ local polynomial weights satisfy $\sum_i \ell_i(x)(X_i - x)^k = \delta_{0,k}$. Deduce that if r is itself a polynomial of degree $\le \ell$, then $\hat{r}_n^{\mathrm{LP}(\ell)}(x) = r(x)$ (zero bias).

Exercise 6 (cross-validation identity). For a linear smoother $\hat{\mathbf{y}} = \mathbf{W}\mathbf{y}$, prove the identity

$$Y_i - \hat{r}_h^{(-i)}(X_i) \;=\; \frac{Y_i - \hat{r}_h(X_i)}{1 - W_{ii}}.$$

Exercise 7 (R, computational). On the `mcycle` data:

- compute Nadaraya–Watson and local linear estimates,
- select the bandwidth by CV and by GCV,
- compare the resulting fits graphically, paying attention to the boundaries.

References

- L. Wasserman, *All of Nonparametric Statistics*, Springer, 2006 (Ch. 5–6).
- A. B. Tsybakov, *Introduction to Nonparametric Estimation*, Springer, 2009 (Ch. 1, local polynomials and Hölder classes).
- J. Fan and I. Gijbels, *Local Polynomial Modelling and Its Applications*, Chapman & Hall, 1996.
- F. Comte, *Estimation non-paramétrique*, lecture notes.
- G. Turinici, lecture notes on nonparametric statistics (kernel and local polynomial regression).
- W. Härdle, *Applied Nonparametric Regression*, Cambridge UP, 1990.